

Chapter 13 Wilson Adiabatic Qualification in Aion

Final NQ9 synthesis, numerical evidence, and Chapter 14 handoff

Aion development record

24 September 2026

Status. NQ0–NQ8 have separate user-accepted review records. The NQ9 synthesis reported here was executed and authenticated, but NQ9 itself is **unreviewed**. No statement in this document changes a mathematical source or extends the accepted physical domain. Raw executions remain immutable; scientific acceptance is carried only by review records.

Repository branch: `feature/ch13-wilson-adiabatic-qualification`.

NQ9 synthesis commit: 74a493241667.

Synthesis result SHA-256:

4eca05f7fbf774a0a47a6d44146929d35941f7fb013acfa6ace973c5080c6671.

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1 Purpose and evidentiary rules

This report is the principal human-readable entry point to the fixed-centre Wilson Hartree and adiabatic Kohn–Sham qualification campaign. It answers the seven questions required by NQ9: which claims have direct numerical support; the systems, bases, fields, functionals, and times over which they were tested; the numerical floor on each claim; the supported reduced electromagnetic action; the observed metric, density-domain, auxiliary, grid, nonlinear, and propagation failures; their basis dependence; and the questions deferred to broader physics.

Four kinds of statement remain distinct throughout:

Mathematical identity

A statement following from the declared continuum action, Wilson frame, and index conventions. It does not assert that code exists.

Reusable implementation

A versioned Aion API. Its presence does not imply that the accepted calculation exercised it.

Executed evidence

A calculation tied to code, environment, fixture, precision, and immutable hashes. Execution alone does not accept the scientific interpretation.

Reviewed evidence

An explicit user decision in `docs/reviews`. Only this state accepts a gate and authorizes the next one.

The independent NQ9 synthesis read the nine NQ0–NQ8 review records, authenticated 41 primary artifacts, flattened 131 accepted measurements and 49 declared limitations, materialized 13 equation-to-code-to-evidence rows, and retained eight explicit negative results. It did not rerun or rewrite any scientific campaign.

2 Physical and mathematical model

2.1 Fixed Wilson frame and density variables

The accepted problem uses fixed nuclei and fixed Gaussian AO anchors $\{\mathbf{R}_i\}$. A straight Wilson path dresses each AO,

$$\chi_i^W(\mathbf{r}, t) = \exp\left[\frac{iq}{\hbar} \int_{\mathbf{R}_i}^{\mathbf{r}} \mathbf{A}(\boldsymbol{\ell}, t) \cdot d\boldsymbol{\ell}\right] \phi_i(\mathbf{r}). \quad (1)$$

The finite-frame overlap and exact Wilson density are

$$S_{ij}(t) = \langle \chi_i^W(t) | \chi_j^W(t) \rangle, \quad (2)$$

$$n_W(\mathbf{r}, t) = P^{ij}(t) \chi_i^W(\mathbf{r}, t) \chi_j^W(\mathbf{r}, t)^*. \quad (3)$$

Here $\mathbf{P} = CfC^\dagger$ is the contravariant coefficient density. The mixed tensor $\mathbf{D} = \mathbf{P}\mathbf{S}$ exposes trace, occupation spectrum, and metric geometry. P and D are not interchangeable in AO contractions.

NQ1 establishes direct versus endpoint/triangle-factorized density agreement, unrestricted real and imaginary coefficient derivatives, fixed-history vector-potential derivatives, magnetic-gauge representative invariance, general complex nonunitary coefficient-frame covariance, and CPU/GPU parity. The largest direct/factorized relative difference is 2.54×10^{-16} ; the finest accepted real-space normalization floor is 2.32×10^{-11} .

2.2 One common nonlinear action

The exact finite-subspace Hartree and adiabatic Kohn–Sham branches use

$$U_H[P, A] = \text{Tr}(PK_{1e}) + E_H[n_W] + E_{NN}, \quad (4)$$

$$U_{KS}[P, A] = \text{Tr}(PK_{1e}) + E_H[n_W] + E_{xc}[n_W] + E_{NN}. \quad (5)$$

The scalar-potential coupling belongs to the temporal connection and is not added a second time to these mechanical energies.

The Coulomb-metric RI realization is a single discrete action,

$$E_H = \frac{1}{2} \rho_A (J^{-1})^{AB} \rho_B, \quad (6)$$

whose lower matrix and fixed-history source derivative are analytic derivatives of the same fitted energy. The selected auxiliary basis is an explicit approximation: its largest accepted deviation from an independent four-centre reference is 6.90×10^{-5} relative. It must therefore be called “RI–Wilson Hartree”, not exact Coulomb.

For LDA, the quadrature action is

$$E_{xc,Q} = \sum_g w_g f(n_W(\mathbf{r}_g)). \quad (7)$$

For GGA it is separately labelled and extended to

$$E_{xc,Q} = \sum_g w_g f(n_W(\mathbf{r}_g), \nabla n_W(\mathbf{r}_g)). \quad (8)$$

Energy, weak lower matrix, and source derivative reuse the same nodes, weights, dressed AO values, dressed gradients, screening rules, and pointwise libxc data. The GGA source derivative includes both $\delta_A n_W$ and $\delta_A \nabla n_W$. LDA evidence is never relabelled as GGA evidence.

2.3 Stationary equation and source observables

At fixed prescribed source, the occupied orbitals obey the nonlinear generalized Hermitian equation

$$K[P]C = SC\varepsilon, \quad (9)$$

with fixed closed-shell occupations. The implementation monitors orbital, density fixed-point, mixed-commutator, metric-orthonormality, particle-number, occupation-spectrum, energy-stationarity, and double-counting residuals.

A vector-source test α differentiates the complete action at fixed coefficient history. The off-shell response separates into minimal, tangential, embedding, and normal-subspace terms. On the coefficient shell, the tangential virtual work vanishes, leaving the action-derived physical weak pairing. No ambient momentum current, graph current, P0 current, or observable from a different approximation is substituted.

The same source construction yields the off-shell Ward identity and the on-shell weak continuity relation

$$\frac{d}{dt} \int f(\mathbf{r}) q n_W(\mathbf{r}, t) d\mathbf{r} = \langle j_{\text{action}}(t), \nabla f \rangle. \quad (10)$$

The constant-weight limit tests global signed-charge conservation. For a Kohn–Sham branch, this is the source current of the declared auxiliary adiabatic action; no exact interacting transverse current is claimed.

2.4 Connection-aware nonlinear propagation

The moving finite Wilson frame supplies the exact overlap, temporal connection, and complete lower mechanical matrix:

$$S\dot{C} = - \left(\omega + \frac{i}{\hbar} K[P, t] \right) C, \quad (11)$$

$$G[P, t] = S^{-1} \left(-\omega - \frac{i}{\hbar} K[P, t] \right), \quad (12)$$

$$\dot{S} = \omega + \omega^\dagger. \quad (13)$$

The connection is evaluated from the prescribed electromagnetic source; it is not reconstructed from $\dot{S}$. Metric compatibility is checked independently.

For each step, the accepted integrator evaluates the nonlinear generator at the two fourth-order Gauss nodes $c_{\pm} = 1/2 \pm \sqrt{3}/6$. Self-consistent node densities define

$$\Omega = \frac{h}{2}(G_- + G_+) - \frac{\sqrt{3}h^2}{12}[G_-, G_+], \quad (14)$$

whose exponential is approximated by the diagonal [2/2] Padé map U . The endpoint update is the congruence

$$P_{n+1} = UP_n U^\dagger, \quad (15)$$

followed by $D_{n+1} = P_{n+1}S_{n+1}$. There is no Löwdin coordinate transformation, Cholesky endpoint correction, metric projection, clipping, or density repair. Cross-metric, trace, and occupation defects remain visible and converge with timestep.

The matrix-rate and action-source power routes independently test

$$U(T) - U(0) = \int_0^T \langle j_{\text{action}}(t), E(t) \rangle dt. \quad (16)$$

A time-dependent uniform magnetic field is always accompanied by its affine induction electric field.

3 Reusable implementation and PySCF boundary

PySCF supplies molecular geometry, all-electron Gaussian bases, bare AO and auxiliary integrals, molecular quadrature grids, AO values and derivatives, field-free RHF/RKS references, and point-wise libxc values. Aion owns the anchors, Wilson phases and gradients, complex pair contractions, exact finite-frame matrices and connection, RI–Wilson action, LDA/GGA action derivatives, non-linear composition, action-source observables, and time propagation.

Table 1: Principal reusable Aion implementation.

Object	Module and responsibility
Wilson density	<code>electronic_structure/wilson_density.py</code> : blocked exact density and matter/source directions
RI Hartree	<code>electronic_structure/ri_wilson_hartree.py</code> : fitted action, lower matrix, and source derivative
LDA action	<code>electronic_structure/wilson_lda.py</code> : common quadrature energy and derivatives
GGA action	<code>electronic_structure/wilson_gga.py</code> : dressed gradients and common PBE action derivatives
Stationary model	<code>electronic_structure/wilson_stationary.py</code> : Hartree/LDA/GGA action composition and generalized SCF
Weak sources	<code>electronic_structure/wilson_sources.py</code> : charge, current decomposition, Ward identity, and continuity
Dynamic model	<code>electronic_structure/wilson_dynamics.py</code> : source samples, nonlinear EOM triple, and power
Reduced actions	<code>electronic_structure/reduced_wilson.py</code> : P0, E1, strict C1, and resummed C1
Time connection	<code>electronic_structure/time_connection.py</code> : exact connection, metric rate, and source directions

Object	Module and responsibility
Propagator	<code>propagation/tensorial.py</code> : self-consistent contravariant-density Gauss–Magnus evolution

GPU execution keeps Wilson frames and matrix contractions device-resident. Only real point-wise libxc inputs and outputs cross the host boundary. The physical-GPU launcher rejects a missing device or silent CPU fallback. NQ8’s complete final GPU regression contained 31 passing tests and one expected CuPy contraction-engine warning.

4 Gate-by-gate evidence

Table 2: Accepted NQ0–NQ8 evidence summarized by NQ9.

Gate	Qualified object	Central evidence	Main limiting floor
NQ0	Contract and field-free reference	H ₂ /STO-3G LDA realization, density/index ledger, component energy reconstruction	4.44×10^{-16} Ha energy closure
NQ1	Exact Wilson density	H ₂ and LiH density, matter/source directions, gauge and frame covariance	2.32×10^{-11} grid normalization
NQ2	RI–Wilson Hartree	Independent four-centre reference; grid, auxiliary, rank, solve, matter, and source scans	6.90×10^{-5} selected auxiliary relative error
NQ3	Wilson LDA	Same-grid PySCF/libxc reference, unrestricted derivatives, gauge/frame covariance	3.27×10^{-9} Ha level-4 to level-5 grid energy
NQ4	Stationary states	H ₂ recovery and H ₃ ⁺ field scan through $B_z = 0.06$	6.53×10^{-9} Ha grid and stationarity floor
NQ5	Sources and continuity	Complete action derivative, Gauss decomposition, off-shell Ward and on-shell weak continuity	1.09×10^{-9} level-5 grid charge
NQ6	Nonlinear dynamics	56 LDA/Hartree trajectories, timestep and nonlinear refinement, power/work identities	6.97×10^{-8} finest density step difference
NQ7	Reduced actions	P0/E1/C1 exact comparisons and three-basis domain stress test	basis-conditioned loss of reduced metric positivity

Gate	Qualified object	Central evidence	Main limiting floor
NQ8	PBE and CO transfer	Same-grid PBE, sources, stationary state, continuity, power, and short dynamics	7.39×10^{-9} CO source derivative

The selected systems are H_2 , LiH , physical two-electron equilateral H_3^+ , and closed-shell CO . Accepted orbital bases are STO-3G and cc-pVDZ; aug-cc-pVDZ appears only in the reduced-domain stress scan, not as a general production qualification. Production fixtures use the weigend auxiliary basis. The qualified closures are Hartree only, restricted pure `lda,vwn`, and restricted pure PBE.

5 Numerical results and error separation

5.1 Action derivatives and stationary states

On exactly matched grids, the LDA energy and lower matrix agree with the independent PySCF/libxc route to 1.33×10^{-15} Ha and 3.39×10^{-16} relative. The PBE counterparts agree to 5.33×10^{-15} Ha and 4.14×10^{-16} relative. These comparisons isolate the XC action from the RI and grid choices.

The accepted H_3^+ stationary LDA scan has maximum orbital and density fixed-point residuals 2.49×10^{-13} and 9.04×10^{-13} . Gauge related energies and real-space densities agree within 2.90×10^{-13} Ha and 8.61×10^{-13} relative. For PBE transfer, the largest field-orbital residual is 3.92×10^{-12} and the minimum metric eigenvalue is 2.17×10^{-2} .

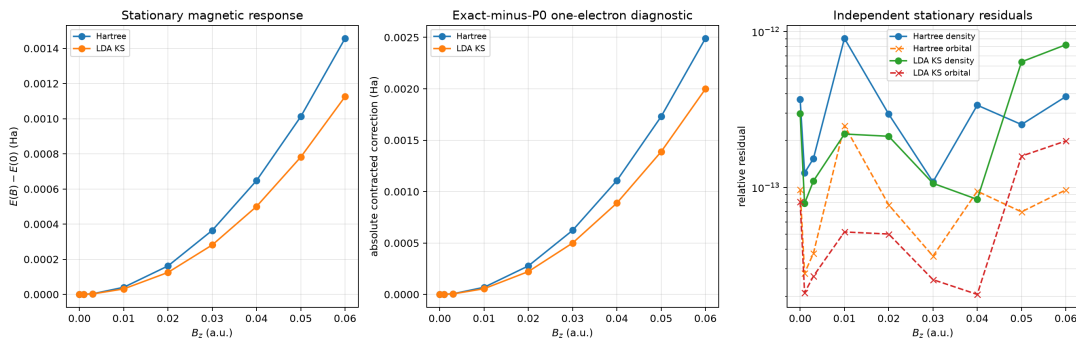


Figure 1: Accepted H_3^+ exact-Wilson stationary field scan from NQ4.

5.2 RI, grid, and finite-difference floors

The Hartree campaign separates real-space quadrature, auxiliary basis, retained rank, and linear solve. The finest selected-grid energy and lower matrix errors are 1.53×10^{-10} and 4.23×10^{-10} relative. The production solve residual is 5.04×10^{-13} , much smaller than the auxiliary approximation. This separation prevents a tight solve from being misreported as exact Coulomb physics.

The complete LDA source derivative closes to 3.64×10^{-10} a.u. in NQ5. The CO/PBE source sequence has a useful central-difference window from 3×10^{-3} to 3×10^{-2} and a best complete residual 7.39×10^{-9} a.u. Finer steps amplify deterministic cancellation in separately evaluated RI-Hartree energies. Repeated transition-region values are bitwise identical, excluding stochastic evaluator noise. The PBE term itself is at its symmetry floor for this CO direction; H_3^+ independently tests a nonzero PBE source contribution.

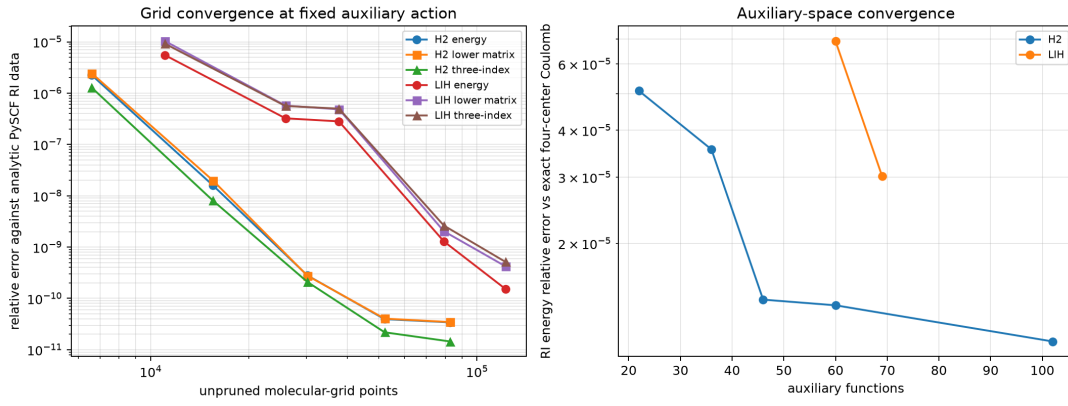


Figure 2: NQ2 grid and auxiliary-basis convergence. The two axes are distinct approximation controls.

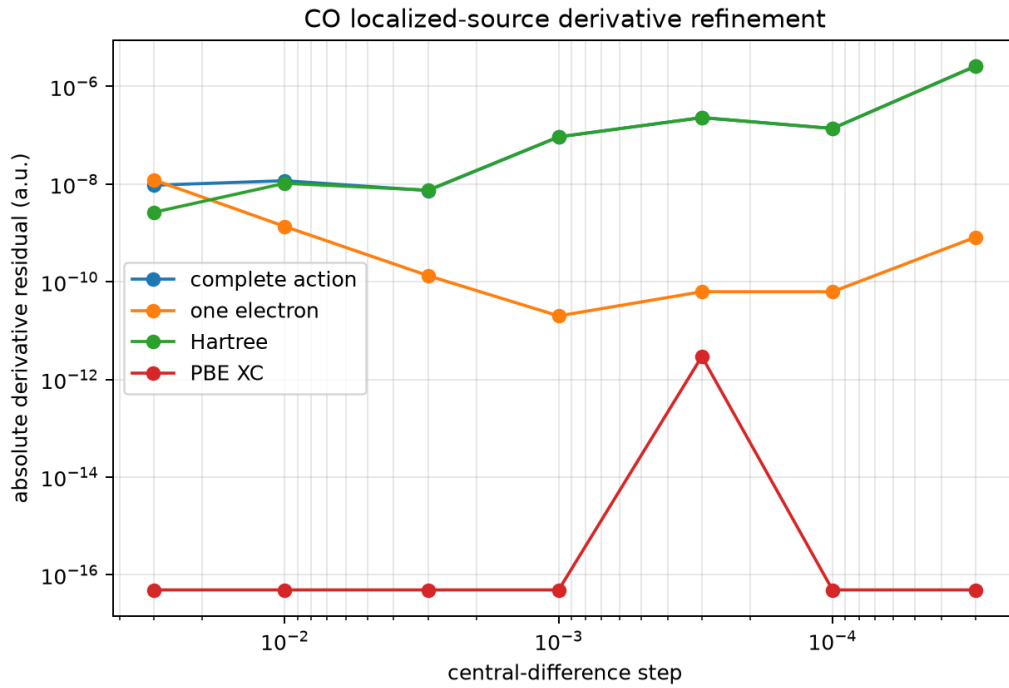


Figure 3: CO localized-source central-difference refinement. The upturn at small step is the resolved RI-Hartree subtraction floor.

5.3 Dynamics, invariants, and power

The NQ6 campaign contains 56 completed trajectories, 920 trajectory rows, and 280 array archives. The minimum measured final-density convergence order is 3.7909. Tightening the nonlinear tolerance changes the final density by at most 7.46×10^{-14} and the energy by 4.94×10^{-13} Ha, separating nonlinear iteration from timestep error.

At the finest step the maximum metric orthonormality, occupation-spectrum, trace, and metric particle-number defects are respectively 1.33×10^{-11} , 1.33×10^{-11} , 2.48×10^{-12} , and 2.66×10^{-11} . The real-space particle-number residual 9.50×10^{-9} is instead limited by the independently measured level-4 spatial quadrature. No metric correction was applied.

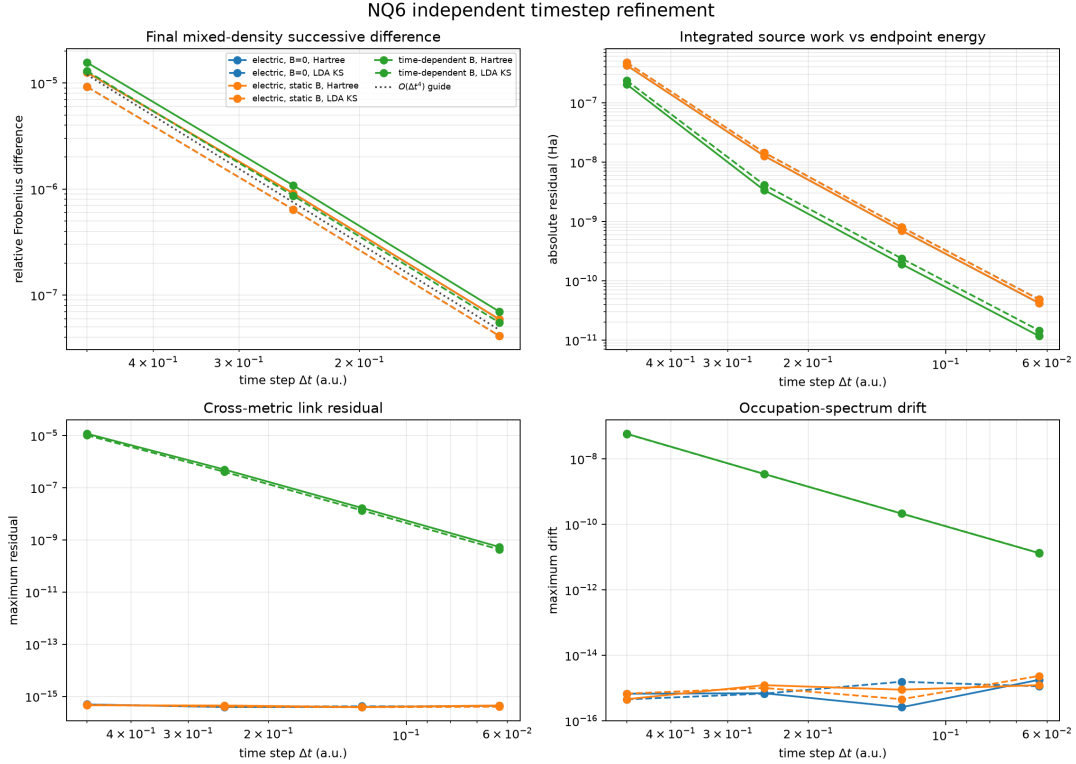


Figure 4: Independent timestep convergence for the accepted nonlinear connection-aware propagator.

The finest instantaneous matrix/source power difference is 3.25×10^{-13} Ha/a.u.; the endpoint mechanical-energy versus integrated source-work defect is 4.87×10^{-11} Ha. The PBE transfer trajectory closes to 4.06×10^{-10} Ha. These are complete-action identities, not comparisons to a current borrowed from another gauge or model.

6 Reduced-model decision and domain failures

All reduced levels retain exact endpoint Wilson links but truncate the endpoint-removed internal finite-spread action differently. P0 deletes the internal spread, E1 adds the first uniform-electric connection increment, strict C1 consistently retains all degree-zero and degree-one terms, and the density-resummed action evaluates unexpanded nonlinear closures on $n_0 + n_1$. The last is a different action, not a repair of strict C1.

At zero magnetic field every reduced level recovers the exact action at the stationary numerical floor. E1 is exact for the tested zero- B uniform electric source: final density, energy trajectory, and power trajectory errors are at most 6.59×10^{-13} , 3.74×10^{-13} , and 1.01×10^{-12} . At nonzero magnetic field, P0 state error is first order while strict C1 is second order. C1 improves stationary

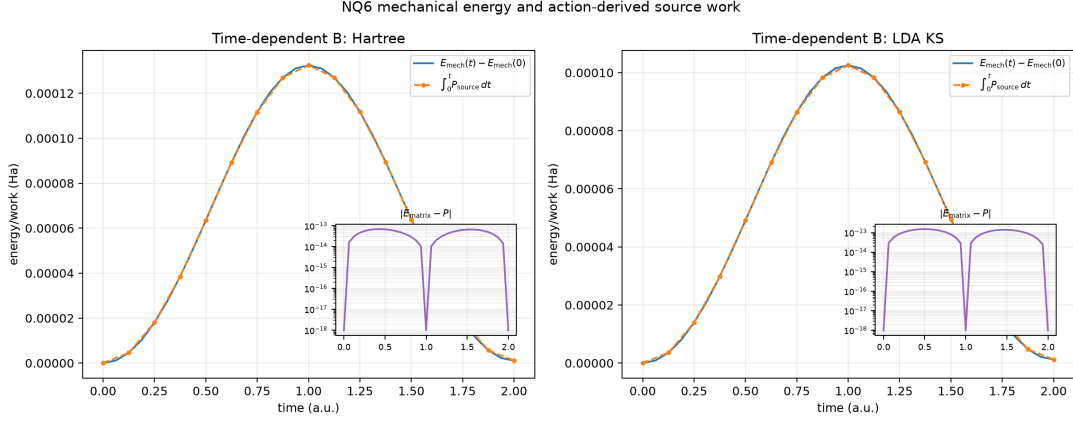


Figure 5: Mechanical energy and integrated source work for the Maxwell-consistent magnetic-induction trajectory.

mixed density by at least a factor 13.43 and dynamic final density over the better of P0/E1 by at least a factor 5.69. The smallest C1 model-error to timestep-error ratio is 6.79×10^3 , so the comparison is physically resolved.

C1 does not uniformly improve every scalar energy or source-power coefficient. This is a retained negative result: a better state and lower matrix do not force every derivative of an omitted second-order action term to improve at the same order. Density-resummed C1 splits the sampled dynamic comparisons evenly with strict C1 and extends no sampled domain. Strict C1 is therefore the accepted near-term reduced action only within this bounded LDA H_3^+ domain.

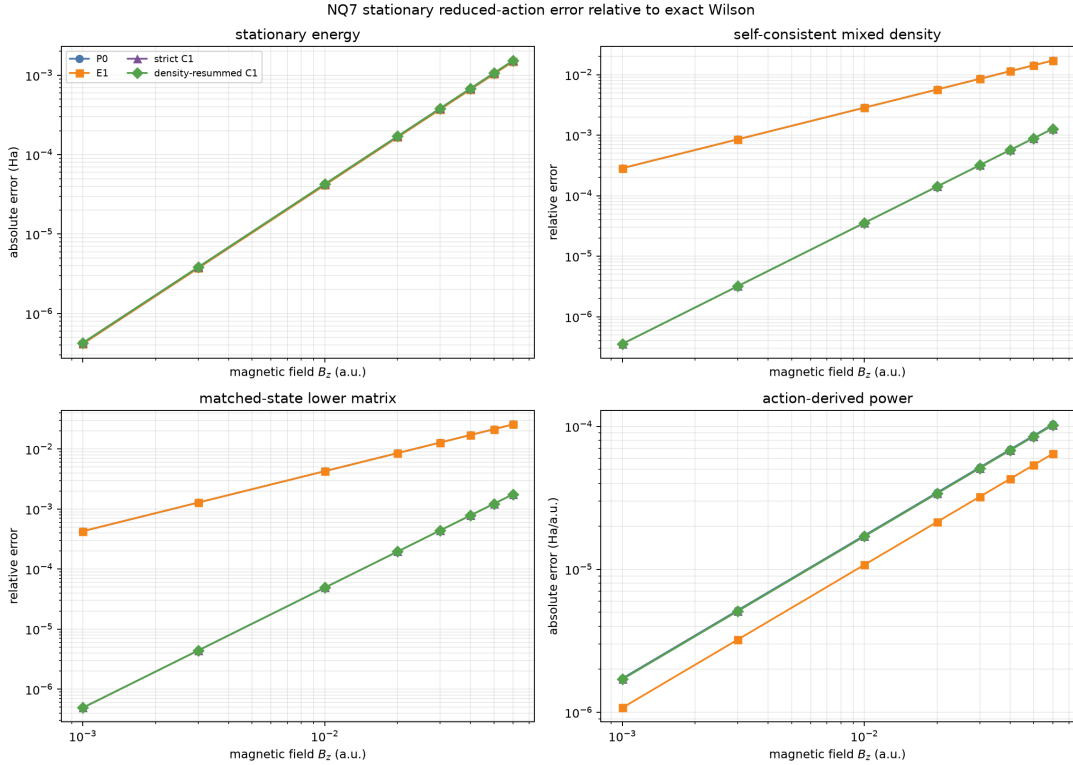


Figure 6: Stationary exact-versus-reduced errors. Strict C1 removes the leading state and lower-matrix error, but not every scalar coefficient.

The decisive domain limitation is basis conditioned. Every tested aug-cc-pVDZ reduced level fails metric positivity at the first nonzero sampled field $B_z = 0.03$ a.u.; the independently recom-

puted exact Wilson metric remains positive at every stress point. Across the full stress scan, 48 reduced failures are visible and none is clipped, regularized, corrected, or projected. The scan brackets failure intervals; it does not locate a continuous critical field and supports no molecule-independent cutoff.

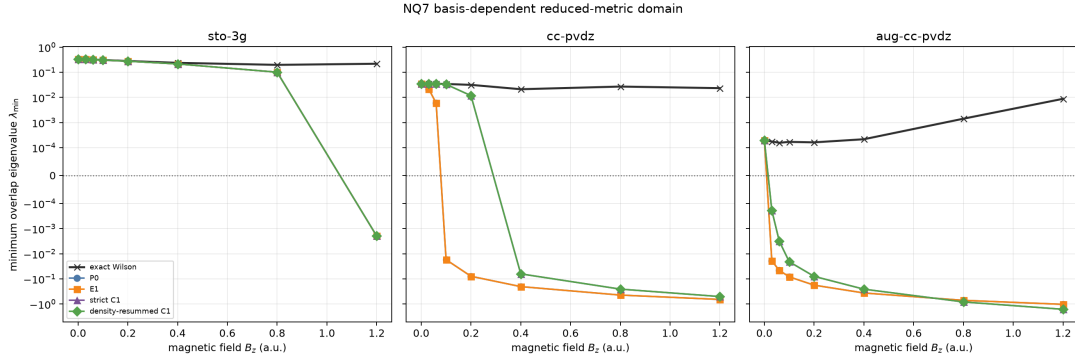


Figure 7: Basis-dependent reduced-metric domain boundaries. The diffuse basis failure is a property of the truncation, not singularity of the exact Wilson overlap.

7 GGA and molecular transfer

NQ8 introduces the separate *quadrature–Wilson GGA* realization. It tests both n_W and ∇n_W responses, same-grid PySCF/libxc agreement, stationary states, sources, continuity, power, and short nonlinear propagation for H_3^+ and closed-shell CO in cc-pVDZ/weigend on unpruned level-4 grids. CO has 28 AOs, 106200 grid points, a peak density 297.701, and maximum density gradient 4521.23, compared with 0.335 and 0.585 for H_3^+ . Nonzero C–O and O–C lower-matrix blocks ensure that the transfer exercises compact core, steep-gradient, and heteronuclear pair regions.

The maximum PBE transfer residuals are 2.37×10^{-13} for weak finite-region continuity, 6.43×10^{-19} for global charge, 3.56×10^{-14} Ha/a.u. for instantaneous power, 4.06×10^{-10} Ha for endpoint work–energy balance, and 2.01×10^{-8} for the final-density difference between timesteps 0.05 and 0.025 a.u. The transfer time is only 0.2 a.u.; it is not a spectroscopy or long-time stability claim. No reduced GGA P0/E1/C1 action is accepted.

8 Negative and failed results

The campaign retains failures as evidence rather than erasing them:

1. the RI auxiliary basis produces a resolved 6.90×10^{-5} relative Hartree approximation error;
2. NQ4 preserves an incomplete campaign, a complete but incorrectly labelled campaign, and a derivative-order fit contaminated by a floor-dominated point;
3. the first NQ5 GPU invocation used the wrong launcher and failed two explicit device-contract assertions before the corrected run passed;
4. a host power failure interrupted 33 of 56 NQ6 trajectories, forcing a clean rerun rather than a splice or undocumented resume;
5. the preliminary NQ6 analysis used continuity and power thresholds below the measured quadrature floor and correctly failed;
6. all sampled aug-cc-pVDZ reduced actions fail their metric domain at the first nonzero field, while exact Wilson remains regular;
7. density-resummed C1 supplies no systematic improvement over strict C1;
8. the CO source derivative reaches an RI energy-subtraction floor; and
9. NQ8 preserves both a shell-appended non-scientific log hash mismatch and a superseded worst-sweep-point component criterion.

The raw numerical NQ8 artifacts authenticate. The final `run.log` changed only because the outer shell appended completion text after the Python process wrote provenance. It is explicitly excluded from scientific evidence rather than silently rehashed.

9 Equation-to-code-to-evidence map

Table 3: Final compact traceability map. Full rows, tests, scopes, and floors are in `tables/equation_code_evidence.csv`.

Object	Mathematical role	Reusable implementation	Gates	Limiting evidence
Density	$n_W = P^{ij} \chi_i^W \chi_j^{W*}$	<code>wilson_density.py</code>	NQ1,3,8	factorization and grid floors
Hartree	$\frac{1}{2} \rho^T J^{-1} \rho$ and derivatives	<code>ri_wilson_hartree.py</code>	NQ2	grid, auxiliary, rank, and solve scans
LDA	$\sum_g w_g f(n_g)$ and derivatives	<code>wilson_lda.py</code>	NQ3	same-grid libxc and directions
GGA	$\sum_g w_g f(n_g, \nabla n_g)$ and derivatives	<code>wilson_gga.py</code>	NQ8	same-grid PBE and CO transfer
Stationary	$K[P]C = SC\varepsilon$	<code>wilson_stationary.py</code>	NQ4,8	residual and gauge scans
Sources	fixed-history action derivative	<code>wilson_sources.py</code>	NQ5,8	source sequences and decomposition
Ward and continuity	gauge identity and weak balance	<code>wilson_sources.py</code>	NQ5,6,8	off-shell and trajectory tests
Connection	$S\dot{C} = -(\omega + iK/\hbar)C$	<code>time_connection.py</code>	NQ6	metric compatibility
Propagation	nonlinear Gauss–Magnus congruence	<code>tensorial.py</code>	NQ6–8	timestep and nonlinear refinement
Power	$\Delta U = \int \langle j, E \rangle dt$	<code>wilson_dynamics.py</code>	NQ6,8	matrix, source, and endpoint work
Reduced models	P0, E1, strict/resummed C1	<code>reduced_wilson.py</code>	NQ7	exact comparisons and domain scan

10 Supported domain and exclusions

The evidence supports one bounded statement:

The exact straight-Wilson finite molecular model, Coulomb-metric RI Hartree closure, restricted pure-LDA and pure-PBE adiabatic actions, their variational lower matrices

and fixed-history weak sources, and connection-aware nonlinear propagation form one numerically consistent implementation over the accepted H_2 , LiH , H_3^+ , and CO fixtures. Strict C1 is a controlled first-order reduced working action inside the sampled pure-LDA H_3^+ domain, whose basis-conditioned failures remain explicit. The exact Wilson action remains the reference.

This statement does *not* qualify moving nuclei, a nonadiabatic nuclear connection, spatially nonuniform propagated fields, Maxwell backreaction, periodic systems, pseudopotentials, spin polarization, hybrids, meta-GGAs, nonlocal correlation, an exact interacting transverse current, reduced GGA actions, long-time stability, spectroscopy, systematic basis convergence, or broad chemical production use.

11 Chapter 14 handoff

The proposed Chapter 14 title is *Numerical realization and qualification of fixed-centre Wilson-projected adiabatic dynamics*. Its recommended sections are:

1. scope, variables, index conventions, and realization tuple;
2. exact Wilson density and variational Hartree/LDA/GGA actions;
3. stationary nonlinear states and gauge/frame covariance;
4. action-derived weak sources, Ward identity, and continuity;
5. connection-aware nonlinear Gauss–Magnus propagation;
6. mechanical energy, source power, and work balance;
7. P0, E1, strict-C1, and resummed-C1 validity and failures;
8. molecular and functional transfer from $\text{H}_2/\text{LiH}/\text{H}_3^+$ to CO/PBE ;
9. numerical error budget, negative results, and reproducibility; and
10. bounded conclusions and open extensions.

The detailed outline is in `docs/chapter13_nq9_chapter14_outline.md`. It is a writing proposal only; NQ9 does not authorize edits to the formalism book.

12 Artifact index and reproduction

The immutable NQ9 synthesis root is

```
/home/cgs/00_WORK/Projection_Code/CALCULATIONS/campaigns/
chapter13_wilson_adiabatic_qualification/
nq9_synthesis_20260924T220919Z_74a493241667.
```

Its entry points are `result.json`, `provenance.json`, and `completed.json`. Derived tables include accepted gates, artifact authentication, every accepted measurement, the full equation/code/evidence map, all review limitations, explicit negative results, and the selected figure manifest.

The nine accepted gate entry points are `docs/chapter13_nq0_contract_and_field_free_reference.md` through `docs/chapter13_nq8_gga_transfer_qualification.md`. Their decisions are in `docs/reviews/chapter13_nq0_review_20260921.json` through `docs/reviews/chapter13_nq8_review_20260924.json`. Those records contain the immutable raw roots and primary hashes.

From a clean repository at commit `74a493241667`, reproduce the synthesis with

```
env OMP_NUM_THREADS=2 OPENBLAS_NUM_THREADS=2 MKL_NUM_THREADS=2 \
NUMEXPR_NUM_THREADS=2 VECLIB_MAXIMUM_THREADS=2 \
/home/cgs/01_TOOLS/EasyBuild/conda/bin/conda run \
-p /home/cgs/01_TOOLS/EasyBuild/conda/envs/aion \
python tools/synthesize_chapter13_nq9.py \
```

```
--repo /home/cgs/00_WORK/Projection_Code/aion \  
--output <new-empty-output-directory>
```

The portable report figures are copied only after their synthesis hashes are verified:

```
python tools/build_chapter13_nq9_report_assets.py \  
--synthesis <nq9-synthesis-root> \  
--output docs/figures/chapter13_nq9
```

The report is compiled with two noninteractive `/usr/bin/pdflatex` passes from `docs`; the compilation log is checked for LaTeX errors, undefined references, and missing files.

13 NQ9 review boundary

The NQ9 synthesis is reconstructive: it finds no missing or mismatched mandatory NQ0–NQ8 artifact and introduces no new physical calculation. It supports archival closure of Chapter 13 and the proposed Chapter 14 writing handoff, subject to explicit user review. Until that review is recorded, NQ9 remains `executed_unreviewed`.